

Explainable Deep Learning for Brain Tumor Classification: A Comparative Study Using MRI Images

Introduction

Brain tumors are among the most life-threatening neurological disorders, characterized by the abnormal growth of cells within the brain. Depending on their type, location, and rate of progression, brain tumors can significantly affect cognitive, sensory, and motor functions and may lead to severe neurological complications or death if not diagnosed and treated at an early stage.

The major categories of brain tumors include:

- Glioma
- Meningioma
- Pituitary Tumor
- No Tumor (Healthy Brain)

Each category exhibits distinct pathological characteristics and requires different treatment procedures, including surgery, radiation therapy, chemotherapy, or continuous monitoring. Consequently, early detection and accurate classification of brain tumors are critical for effective treatment planning and improving patient survival rates.

Magnetic Resonance Imaging (MRI) has become the most widely used imaging modality for diagnosing brain abnormalities because it provides high-resolution images of soft tissues without exposing patients to ionizing radiation. MRI scans provide valuable information regarding the size, shape, texture, and location of tumors.

However, interpreting MRI images remains a complex and challenging process.

Background of the Problem

Modern healthcare facilities generate a massive amount of medical imaging data every day. A single MRI examination may consist of hundreds of image slices that require careful analysis and interpretation by expert radiologists.

The diagnosis and classification of brain tumors using MRI images involve several challenges:

Time-Consuming Process

Radiologists must manually inspect numerous MRI slices to determine the presence and type of tumor. This process requires significant expertise and may delay diagnosis in high-volume healthcare environments.

Dependency on Specialized Experts

Accurate brain tumor diagnosis relies heavily on experienced neuroradiologists. Many hospitals, particularly those in rural and under-resourced regions, lack sufficient medical specialists, resulting in delayed diagnosis and treatment.

Human Variability

Interpretation of MRI images often varies between radiologists because of differences in experience and subjective judgment. The same MRI scan may receive different diagnoses from different specialists.

Diagnostic Complexity

Brain tumors exhibit considerable variations in:

- Shape
- Size
- Intensity
- Texture
- Position

Furthermore, certain tumors share similar imaging characteristics, making differentiation difficult even for experienced clinicians.

Increasing Medical Workload

The growing incidence of neurological disorders and increasing demand for diagnostic imaging have significantly increased the workload on healthcare professionals, creating a need for intelligent computer-aided diagnostic systems.

Research Problem

Recent advancements in Artificial Intelligence and Deep Learning have demonstrated remarkable success in medical image analysis. Deep learning models can automatically learn hierarchical features from MRI images and assist radiologists in disease detection and classification.

Although numerous studies have proposed deep learning solutions for brain tumor classification, several challenges remain unresolved.

Challenge 1: Lack of Reliable Comparative Evaluation

Most studies employ a single architecture or compare models under different experimental conditions, making it difficult to determine which architecture performs consistently better for brain tumor classification.

Challenge 2: Computational Efficiency

Medical imaging applications require models that are not only accurate but also computationally efficient and scalable for real-world deployment.

Challenge 3: Black-Box Nature of Deep Learning

Most deep learning systems provide predictions without explaining how decisions were made. Healthcare professionals require transparent and interpretable systems before incorporating artificial intelligence into clinical practice.

Challenge 4: Clinical Trust and Adoption

Even highly accurate models may not be accepted in healthcare environments if clinicians cannot understand or verify the decision-making process.

These limitations demonstrate the need for an intelligent, explainable, and comparatively evaluated deep learning framework specifically designed for MRI-based brain tumor classification.

Statement of the Problem

Brain tumor diagnosis from MRI images remains a challenging and time-sensitive task due to the complexity of medical images, dependence on expert interpretation, and increasing diagnostic workload. Manual analysis of MRI scans is often time-consuming, susceptible to human variability, and constrained by the availability of specialized radiologists.

Although deep learning techniques have shown significant potential in medical image classification, there is insufficient comparative research evaluating the effectiveness of advanced architectures such as ResNet50 and EfficientNet for brain tumor classification under a unified experimental framework. Furthermore, many existing systems operate as black-box models and fail to provide interpretable explanations that are essential for clinical acceptance.

Therefore, there exists a need to develop an explainable deep learning framework capable of automatically classifying brain tumors from MRI images while systematically comparing ResNet50 and multiple EfficientNet variants (B0–B7) to determine the most accurate, computationally efficient, and clinically interpretable architecture.

Proposed Research Solution

This research proposes an Explainable Deep Learning Framework for Brain Tumor Classification Using MRI Images.

The framework consists of the following stages:

Stage 1: Dataset Preparation

MRI images containing the following classes will be collected:

- Glioma
- Meningioma
- Pituitary Tumor
- No Tumor

The dataset will be divided into:

- Training Set
- Testing Set

Stage 2: Image Preprocessing

MRI images will undergo preprocessing procedures including:

- Image resizing
- Normalization
- Data cleaning
- Standardization

Image preprocessing improves data quality and ensures consistent input for model training.

Stage 3: Deep Learning Model Development

The study will implement and evaluate:

ResNet50

ResNet50 utilizes residual learning and skip connections to enable deeper neural networks while mitigating the vanishing gradient problem.

Advantages:

- Excellent feature extraction capability
- Strong performance in medical imaging tasks
- Effective learning of complex pattern.

EfficientNet (B0–B7)

EfficientNet introduces compound scaling by simultaneously optimizing:

- Network depth
- Network width
- Input image resolution

Advantages:

- High classification accuracy
- Lower computational complexity
- Better parameter efficiency
- Scalability across model variants

The study will investigate multiple EfficientNet variants ranging from B0 to B7 and compare their performance with ResNet50.

Stage 4: Explainable Artificial Intelligence

Since medical applications demand transparency, the study will incorporate:

Grad-CAM (Gradient-weighted Class Activation Mapping)

Grad-CAM generates heat maps highlighting regions of MRI images that contribute most significantly to model predictions.

The integration of Grad-CAM aims to:

- Improve model transparency
- Increase clinician confidence
- Validate whether the model focuses on actual tumor regions
- Support trustworthy clinical decision-making

Research Objectives

The objectives of this research are:

Objective 1

Develop an automated MRI-based brain tumor classification framework using deep learning.

Objective 2

Implement and compare ResNet50 and EfficientNet architectures (B0–B7).

Objective 3

Evaluate model performance using:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC
- Confusion Matrix

Objective 4

Incorporate Grad-CAM to generate visual explanations for predictions.

Objective 5

Identify the most accurate, efficient, and explainable architecture for MRI-based brain tumor classification.

Expected Outcomes

The proposed research is expected to:

- Develop a robust automated brain tumor classification system.
- Determine whether ResNet50 or EfficientNet variants provide superior performance.
- Generate explainable predictions through Grad-CAM visualizations.
- Reduce diagnostic workload for radiologists.
- Improve confidence in AI-assisted medical diagnosis.
- Provide a reproducible and scalable framework for future medical imaging research.

Significance of the Research

Clinical Significance

- Supports radiologists during diagnosis.
- Reduces interpretation time.
- Assists in early tumor detection.

Technological Significance

- Provides a comparative benchmark between ResNet50 and EfficientNet architectures.
- Demonstrates the application of Explainable AI in medical imaging.

Research Significance

- Addresses the lack of comparative evaluation under a standardized framework.
- Bridges the gap between high classification accuracy and model interpretability.
- Contributes toward trustworthy and clinically deployable artificial intelligence systems.

Final Problem Statement

The increasing complexity of MRI-based brain tumor diagnosis, combined with dependence on expert interpretation, diagnostic variability, and the lack of transparent automated systems, necessitates the development of an explainable deep learning framework for brain tumor classification. This research therefore proposes a comparative evaluation of ResNet50 and EfficientNet (B0–B7) architectures integrated with Grad-CAM-based explainability to identify the most accurate, computationally efficient, and clinically interpretable model for MRI-based brain tumor diagnosis and decision support.